5 **(a)** Define *electric potential* in an electric field.

.....

..... [1]

(b) Two spherical charges P and Q, each of mass m = 0.20 g are separated by a horizontal distance of 30 mm as shown in Fig. 5.1. P has a charge of +20 μC and Q has a charge of –40 μC.

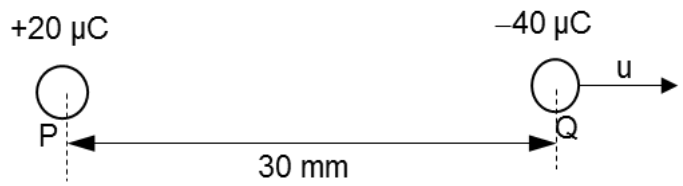

Fig 5.1

(i) In Fig 5.2, r represents the distance from centre of P towards centre of Q. On the axis provided, sketch the variation of the resultant electric potential V between boundaries A and B.

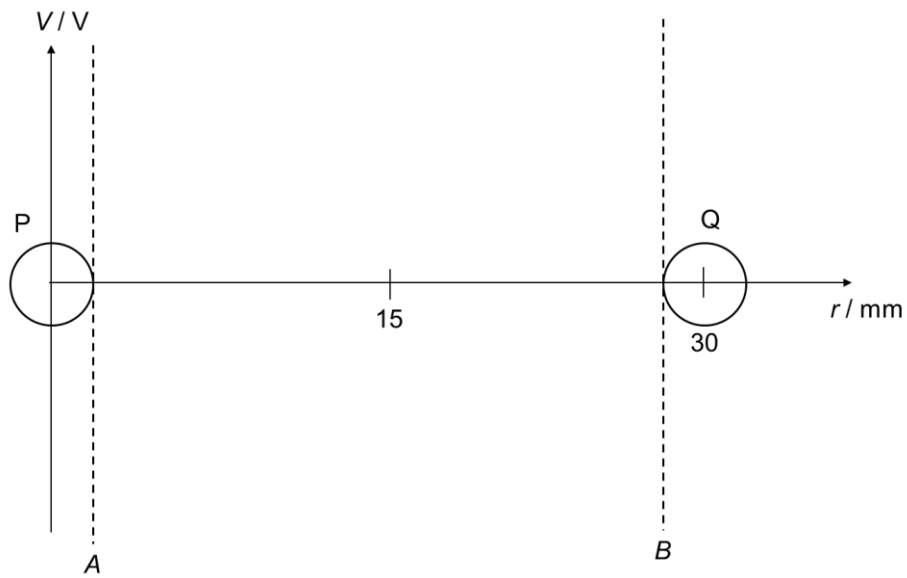

Fig 5.2

[3]

(ii) Q is then projected away from P with an initial speed *u* such that it reaches a position far enough beyond the influence of charge P.

1. Determine the electric potential energy of Q at its initial position.

electric potential energy = …….. J [2]

2. Determine the minimum initial speed *u* required.

minimum speed *u* = ……. m s^{-1} [2]